

Flood Risk Assessment

White Rock Creek Watershed

Dallas County, Texas

A HEC-RAS 2D Hydraulic Modeling & Exposure Analysis

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Contents

Contents 2

1. Executive Summary 3

2. Introduction & Study Area 3

3. Methodology 3

 3.1 Data Acquisition..... 3

 3.2 Terrain & Watershed Delineation 4

 3.3 Hydraulic Modeling 4

 3.4 Exposure & Risk Analysis 4

4. Results..... 5

 4.1 Inundation by Recurrence Interval 5

 4.2 Modeled vs FEMA Regulatory Floodplain 5

 4.3 Exposure Analysis (100-Year Event) 5

 4.4 Social Vulnerability 6

5. Limitations 6

6. Conclusions 6

1. Executive Summary

This assessment characterizes flood hazard and its consequences across the White Rock Creek watershed, an 85.5-square-mile, largely urbanized basin in northeast Dallas County, Texas. Using two-dimensional hydraulic modeling in HEC-RAS 7.0.1, the study simulated four design storms — the 10-, 50-, 100-, and 500-year events — and intersected the resulting floodplains with buildings, population, transportation, critical facilities, and census-based social vulnerability.

The modeled 100-year (1% annual chance) flood inundates 13,480 acres and exposes an estimated 69,913 residents (13.8% of the watershed population), 3,766 buildings, and 398 miles of roadway. Most consequentially, two of the watershed's eleven hospitals fall within the modeled 100-year floodplain, and eight socially vulnerable census block groups have more than 10% of their area exposed — indicating that flood risk is not distributed equally across the community.

The modeled floodplain exceeds FEMA's regulatory mapping by 115%, a difference attributable to the rain-on-grid method's capture of pluvial and tributary flooding beyond FEMA's riverine study corridors. Where FEMA has performed detailed study, the model is in close agreement.

2. Introduction & Study Area

White Rock Creek drains approximately 85.5 square miles (54,696 acres) of predominantly developed land before discharging into White Rock Lake and continuing to its confluence with the Trinity River. The watershed encompasses portions of Dallas, Garland, and surrounding communities, with a 2024 population of approximately 508,251.

The objective of this assessment is to quantify flood hazard across a range of recurrence intervals and to translate that hazard into exposure and risk metrics meaningful for emergency management, mitigation planning, and equitable resource allocation.

Watershed Characteristic	Value
Drainage area	85.5 sq mi (54,696 ac)
Population (2024 ACS)	508,251
Primary waterbody	White Rock Lake
Hydrologic unit	Trinity Basin (12030105)
Outlet	Confluence with Trinity River
Setting	Urban / suburban

3. Methodology

3.1 Data Acquisition

Source datasets were assembled from authoritative public providers and projected to NAD83 State Plane Texas North Central (EPSG 2276):

- Terrain: USGS 3DEP digital elevation model
- Hydrography: USGS National Hydrography Dataset (NHD)
- Land cover: USGS National Land Cover Database (NLCD)

- Precipitation: NOAA Atlas 14 precipitation-frequency estimates
- Demographics: US Census Bureau ACS 2024 5-Year Estimates
- Critical facilities: Esri Living Atlas (hospitals, fire, police, schools)
- Buildings: Dallas County footprint centroids
- Regulatory floodplain: FEMA National Flood Hazard Layer (NFHL)

3.2 Terrain & Watershed Delineation

The DEM was hydrologically conditioned (fill), and flow direction and accumulation were computed to delineate the contributing watershed from an automatically detected pour point at the creek outlet. The resulting 85.5-square-mile boundary defined the modeling domain.

3.3 Hydraulic Modeling

Flooding was simulated using a two-dimensional, rain-on-grid (direct rainfall) approach in HEC-RAS 7.0.1. NOAA Atlas 14 design-storm depths were distributed across a 24-hour SCS Type II temporal pattern and applied uniformly to a 287,909-cell computational mesh. The Diffusion Wave equation set was selected after confirming gradually-varied flow conditions; the final 100-year simulation achieved a volume-accounting (mass-balance) error of 0.0085%.

Model Parameter	Value
Software	HEC-RAS 7.0.1 (2D)
Method	Rain-on-grid (direct precipitation)
Equation set	Diffusion Wave
Mesh resolution	287,909 cells (~100 ft)
Computation interval	30 seconds
Simulation window	48 hours
Mass-balance error	0.0085%

Design-storm depths at the watershed centroid:

Recurrence Interval	24-hour Depth (in)
10-year (10% AEP)	6.03
50-year (2% AEP)	8.42
100-year (1% AEP)	9.55
500-year (0.2% AEP)	12.60

A 1.0-foot depth threshold was applied to reported flood extents to distinguish meaningful flooding from shallow surface ponding inherent to the rain-on-grid method.

3.4 Exposure & Risk Analysis

Modeled flood extents were intersected with building, population, road, and critical-facility datasets. Population exposure was estimated by areal weighting of census block groups. A Social Vulnerability Index (SVI) was computed from poverty rate, vehicle access, and elderly population, standardized to z-scores and classified into Low, Moderate, and High categories. A

weighted flood-risk index combined normalized depth (40%), velocity (25%), population (20%), and infrastructure proximity (15%) to support mitigation prioritization.

4. Results

4.1 Inundation by Recurrence Interval

Modeled inundation increased monotonically with storm severity, confirming internal consistency across the four events:

Event	Inundated Area (ac)	Max Depth (ft)	Max Velocity (fps)
10-year	9,868	21.8	21.5
50-year	12,290	24.5	24.7
100-year	13,480	25.6	24.8
500-year	16,162	28.2	25.2

4.2 Modeled vs FEMA Regulatory Floodplain

The modeled 100-year floodplain (13,480 ac) exceeds FEMA's regulatory 100-year floodplain (6,276 ac, comprising A, AE, and AO zones) by 7,204 acres, or 115%. FEMA's mapping in this watershed is almost entirely detailed-study (AE) along main channels. The additional modeled extent reflects pluvial and tributary flooding captured by the rain-on-grid approach but outside FEMA's riverine study scope. This is a difference of method coverage, not model error; where FEMA has studied in detail, the model agrees.

100-Year Floodplain	Area (acres)
FEMA regulatory (A/AE/AO)	6,276
HEC-RAS 2D modeled	13,480
Difference	+7,204 (+115%)

4.3 Exposure Analysis (100-Year Event)

Asset	Exposed	Total	Percent
Buildings	3,766	110,168	3.4%
Population	69,913	508,251	13.8%
Road miles	398	2,463	16.2%
Hospitals	2	11	18%
Schools	4	101	4%
Fire stations	0	16	0%
Police stations	0	5	0%

The exposure of two hospitals (18% of the watershed total) is the most operationally significant finding, with direct implications for emergency access and continuity of care during major flood events.

4.4 Social Vulnerability

Eight census block groups classified as High social vulnerability have more than 10% of their area within the modeled 100-year floodplain. This spatial coincidence of elevated vulnerability and flood exposure identifies priority communities where flooding is most likely to produce severe and inequitable impacts, and where mitigation and outreach investment would yield the greatest equity benefit.

5. Limitations

- The rain-on-grid method applies spatially uniform rainfall and does not route external inflows entering from upstream of the watershed boundary.
- Default Manning's roughness was applied where land-cover-specific values were unavailable.
- Reported flood extents reflect a 1.0-foot depth threshold; results are sensitive to this choice.
- Building exposure is evaluated at point (centroid) locations and does not model partial structural flooding.
- Population exposure uses areal weighting and assumes uniform population distribution within each block group.
- Critical-facility locations derive from Esri Living Atlas and were not independently field-verified.

6. Conclusions

The White Rock Creek watershed faces substantial and spatially concentrated flood risk. A 100-year event would inundate more than 13,000 acres, expose nearly 70,000 residents, and place two hospitals within the floodplain. Flood exposure disproportionately affects socially vulnerable communities, with eight high-vulnerability block groups significantly impacted.

The two-dimensional rain-on-grid model provides a more complete hazard picture than channel-only regulatory mapping, capturing pluvial and tributary flooding that conventional methods omit while remaining consistent with FEMA where detailed study exists. These results support targeted mitigation: protecting critical healthcare infrastructure, prioritizing flood-prone vulnerable communities, and incorporating pluvial flood pathways into local planning.

This assessment demonstrates an end-to-end geospatial workflow — from terrain conditioning and 2D hydraulic modeling through exposure, equity, and risk analysis — applicable to flood-risk evaluation in any urbanizing watershed.